

A survey of large language models for cyber threat detection[☆]

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ABSTRACT

With the increasing complexity of cyber threats and the expanding scope of cyberspace, there exist progressively more challenges in cyber threat detection. It is proven that most previous threat detection models may become inadequate due to the escalation of hacker attacks. However, recent research has shown that some of these problems can be effectively addressed by Large Language Models (LLMs) directly or indirectly. Nowadays, a growing number of security researchers are adopting LLMs for analyzing various cyber threats. According to the investigation, we found that while there are numerous emerging reviews on the utilization of LLMs in some fields of cyber security, there is currently a lack of a comprehensive review on the application of LLMs in the threat detection stage. Through retrieving and collating existing works in recent years, we examined various threat detection and monitoring tasks for which LLMs may be well-suited, including cyber threat intelligence, phishing email detection, threat prediction, logs analysis, and so on. Additionally, the review explored the specific stages of different detection tasks in which LLMs are involved, evaluating the points at which LLMs are optimized. For instance, LLMs have been found to enhance the interpretability of log analysis in real-time anomaly event discovery. Additionally, we discussed some tasks where LLMs may not be suitable and explored future directions and challenges in this field. By providing a detailed status update and comprehensive insights, this review aims to assist security researchers in leveraging LLMs to enhance existing detection frameworks or develop domain-specific LLMs.

1. Introduction

Over the past several decades, the landscape of cyberspace threats has witnessed a progressive escalation in both frequency and complexity, culminating in widespread security incidents and substantial economic losses globally. This trend has unfolded in parallel with the evolution of tactics, techniques, procedures (TTPs), and strategies between hackers and security researchers (Kaur et al., 2023). Among various cyberspace threats, the Advanced Persistent Threat (APT) has emerged as a significant contributors to the security dilemmas faced by governments and corporations alike. For example, the APT organization FamousSparrow targeted key sectors across multiple Asian and African countries in 2021, causing serious chaos to telecommunications infrastructure. Besides, more and more similar security incidents have been continuously reported recently, such as Colonial Pipeline ransomware

attacks in 2021, Kaseya VSA supply chain attacks in 2021, Microsoft Exchange Server events in 2021, etc. Resulting of various APTs, the situation of cyberspace security came across an unprecedented crisis.

As threats continue to evolve in their persistence, covert nature, and unpredictability, conventional threat detection systems are facing challenges in effectively detecting novel threats. Rule-based, machine-learning, and deep-learning models are encountering difficulties in keeping pace with the dynamic and evolving landscape of threat detection and discovery (Yuan and Wu, 2021). This challenge is predominantly attributed to the proliferation of threat sources, the increasing intricacy of threat compositions, and the broadening reach of cyberspace. Concurrently, cyber threats emanate not solely from vulnerabilities in information systems and malicious activities of hackers, but also from various other factors including internal threats, espionage,

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errors in security management, and zero-day exploits, among others. Despite numerous efforts (Yaacoub et al., 2020) to improve automation and intelligence in threat detection and discovery, security researchers are still faced with the challenge of effectively managing large amounts of network data and complex security tasks. Key issues include the limited scalability of detection systems, high rates of false alarms in real-world security situations, and susceptibility to adversarial attacks. Researchers increasingly recognize that addressing these challenges necessitates reassessing existing methodologies and seeking more robust, adaptable, and intelligent solutions in the realm of cyber threat detection.

Given this, some remarkable discoveries have suggested that Large Language Models (LLMs), particularly GPT and BERT developed in recent years, are capable of providing more effective solutions for some of these issues. Not only have LLMs shown superior performance in large-scale data works, multi-modal tasks content generation, etc., but can simultaneously support functions of perception, cognition, and decision-making in artificial intelligence (AI) tasks (Bommasani et al., 2021). As a result, there has been a surge in research utilizing LLMs for cyberspace threat detection and discovery, with these models rapidly gaining traction among researchers. According to NIST2.0, cyber security works can be roughly divided into five phases: identification, protection, detection, response, and recovery. Although numerous systematic reviews have examined the use of LLMs in some fields of cyber security (Motlagh et al., 2024; Xu et al., 2024; Sarker, 2024), there remain few special discussions for the integration of LLMs with cyber threat detection stage. Therefore, we compiled this review to summarize some representative applications of LLMs in cyberspace threat detection and discovery, which may assist security researchers in efficiently navigating this emerging field.

Compared to other similar works, this review has some unique contributions, as shown below:

- This review provides a relatively exhaustive survey of LLMs utilized specifically to detect cyber threats, focusing on thematic works that can be applied in practical security scenarios. From the perspective of defenders, we focus on exploring how to use LLMs to enhance the detection of cyber threats.
- The research has categorized various stages of tasks in security scenarios that are closely correlated to LLMs. In this way, can we investigate the suitability of LLMs for specific security tasks and their potential roles across different stages. This review seeks to support the smooth incorporation of LLMs into existing security frameworks or the creation of domain-specific LLMs by security experts from diverse research fields.
- After comparing thematic works, We have compiled further analyses on LLMs and cyber threats from various scholarly sources and publications, synthesizing and presenting our perspectives on the potential advantages and obstacles of utilizing LLMs for detecting cyber threats and enhancing overall network security.

The rest of the review is organized as follows: Section 2 delves into a brief history of LLMs and threat detection, finding interconnections between them. In Section 3, the majority of related works about applications of LLM on threat detection are presented in different security scenarios. In Section 4, we introduced the methodology followed for this review and further analyzed what we found. In Section 5, we shared extra opinions on related topics. In the end, we concluded this study and pointed out some future research directions and challenges in Section 6.

2. Background

The combination of AI and cyberspace threat detection dates back to the 1990s when security researchers attempted to apply Neural Networks (NNs) in Intrusion Detection Systems (IDS). From then on, an

increasing number of researchers have employed AI methods to address security challenges. With the advancement of deep learning (DL), LLMs were born in the field of Natural Language Processing (NLP). After that, some applications of LLMs have emerged in cybersecurity, notably in threat detection (Gupta et al., 2023). In light of this, this section aims to outline some key milestones in the history of LLMs and cyberspace threat detection, shedding light on the inherent connections between these two domains.

2.1. Large language models

With the scale of NNs expanding, researchers were delighted to find that DL models have begun to exhibit a series of advanced abilities that traditional models cannot demonstrate. These advanced abilities involve a chain of thought reasoning, human sentiment analysis, complex context abstracting, etc. This phenomenon, known as Intellectual Emergence (Zoph et al., 2022), serves as a crucial distinguishing factor between large models and their predecessors. One type of such large model, the Language Model (LLM), is predominantly utilized in natural language processing (NLP) applications and has demonstrated proficiency in various language-related tasks, such as text generation, question and answer (Q&A), intelligent retrieval, etc.

To achieve this, specialized technical designs must be developed for LLMs. On one hand, billions of parameters and efficient network structures like Transformer (Vaswani et al., 2017) are required to improve the expressive capabilities of models. GPT-3, for example, is reported to contain over 175 billion parameters. On the other hand, more high-quality training data plays a crucial role in enabling LLMs to acquire a wider and more diverse range of knowledge, improving their generalization ability in different tasks. According to Brown et al. (2020), the training dataset utilized for GPT-3 comprises 45 terabytes of compressed plain text data, resulting in approximately 570 GB after filtering. Furthermore, the efficacy of LLMs is also attributed to advanced training methods, robust computational resources, and other contributing factors.

Nowadays, alongside the widely used GPT model, there exist numerous LLMs from other institutions such as the open-source PaLM2, the superior multimodal and multitasks LLM Gemini from Google, the advanced Llama from Meta, etc. Most of these LLMs have demonstrated notable applications in fields of finance, healthcare, education, customer service, etc. To realize LLMs better, it is imperative to consider significant advancements in the historical development of these models.

The origin of LLMs can be traced back to early NLP models, particularly the Word2Vec method designed by Mikolov et al. (2013). This method addressed the limitations of semantic mining in previous models by embedding words into a continuous vector space using shallow neural networks. By mapping semantically similar words to adjacent positions in the vector space, the model can predict other words based on context or target words, capturing hidden semantic features. It was Word2Vec that first proposed the idea of text processing by unsupervised learning, which also laid the foundation for the emergence of LLM such as BERT, GPT, etc. Meanwhile, Cho et al. (2014) conducted optimization of Long-Short-Term-Memory (LSTM) networks by introducing the Gated Recurrent Unit (GRU) to enhance the learning capabilities of language models for long-distance dependencies, a development that holds significant implications for modeling sequences in LLMs.

As time went by, more high-quality data and computing resources became available. AI researchers started to be concerned about transfer learning and Pre-Training models. As a result, there emerged two groundbreaking language models in 2018: BERT by Google and GPT by Open AI. Devlin et al. (2019) constructed BERT adopting a bidirectional Transformer architecture, allowing for the simultaneous extraction of contextual information from both directions of word vectors. It was demonstrated that BERT can learn universal language patterns and assess contextual relevance through pre-training on extensive datasets,

Table 1

The comparison of two prevailing style LLMs.

Basic architecture	Main features	Representative models
BERT - style (Encoder–Decoder or Encoder-Only)	Masked pretraining Rely on one-way context Judgmental model	ELMo, BERT, Roberta, DistilBERT, BioBERT,XLM,Xlnet,ALBERT,ELECTRA,T5, XLM-E,ST-MoE,AlexaTM
GPT - Style (Decoder-Only)	Auto-regressive pretraining Rely on two-way context Generative model	GPT-3, OPT, PaLM, BLOOM, GLM, MT-NLG, GLaM, Gopher, chinchilla, LaMDA, GPT-J, Llama, GPT-4, BloombergGPT

followed by fine-tuning for specific tasks. Consequently, BERT has demonstrated exceptional performance across various tasks, including expert Q&A, text classification, and named entity recognition (NER). And (Radford and Narasimhan, 2018) chose the auto-regressive Transformer for GPT-1, which boasts over 117 million parameters. Through a combination of unsupervised pre-training and supervised fine-tuning, GPT has shown proficiency in handling complex NLP tasks, particularly in text generation. In reviewing the history of LLM, can we find that BERT and GPT have jointly played milestone roles as well as represented two prominent pathways for subsequent LLM developments, as illustrated in Table 1.

As mentioned above, BERT and GPT have set a widely accepted performance standard for new language models. Just four months later, Open AI successfully developed GPT-2 in 2019. and then went on to launch the groundbreaking GPT-3 in 2020. Subsequently, Open AI unveiled InstructGPT in March 2022. Notably, InstructGPT uses reinforcement learning with human feedback (RLHF) to fine-tune the large language model to produce output more in line with human preferences. By this means, GPT has been competent in multitasking learning, inference, prediction, and zero-shot learning. Simultaneously, there has developed a series of improved BERTs such as RoBERTa (Zhuang et al., 2021), DeBERTa (He et al., 2020), applied in text processing, entity recognition, and intelligent translation, etc. However, people started to worry about various sectors of the security and ethical issues of LLMs at the same time. Over time, different LLM-driven products especially the ChatGPT have achieved technological and commercial success in different domains by 2023, pushing the LLMs to the forefront of AI capital. More and more LLMs are being developed, serving users with higher-quality content generation, multi-modal processing, deep semantic analysis, etc. Now, it has appeared a worldwide scene named the hundred-models War describing this phenomenon. Concurrently, there are more products like GPTs trying to help users build their own GPT requiring less or even no coding work. This work can greatly reduce the threshold for people to use LLM but also catches our attention to AI security and privacy protection.

Given these works, We can learn something about the development status of LLMs. Firstly, It is evident that DL models have witnessed a dramatic shift from quantity to quality, leading to an unexpected miracle. Although most of them still use the Transformer or other options as the model's fundamental unit, there exists a tendency for more parameters, more layers, and more input data in LLMs. Besides, The open-source LLMs and closed-source LLMs can collaborate and promote each other. Several proprietary LLMs have been improved utilizing open-source frameworks, such as Llama-2, and subsequently enhanced through the creation of original concepts and datasets. The continuous progress of LLM technology is centered on refining encoders and decoders, marking a substantial advancement. These improvements are predominantly evident in novel applications through pre-training and fine-tuning procedures. Additionally, various domain-specific LLMs, such as SecurityGPT, have emerged as a result of this developmental trajectory.

Despite the great reputation of LLMs in certain aspects, there are still some considerable limitations in LLMs. Firstly, Issues such as prompt injection, data poisoning attacks, and backdoor attacks can sensibly decrease the quality of output and the model's efficiency. Additionally, threats such as model theft and reconstruction can compromise the integrity and usability of LLMs. Fortunately, most of such problems

Table 2

Examples of external and internal threats.

Threats category	Examples
External threats	Malware, Phishing attacks, Ransomware, Distributed Denial of Service (DDoS), Viruses, Trojans, Zero-Day Exploits, Spyware, APTs, Network Intrusion, Exploit Kits, etc.
Internal threats	Users' Negligence (misconfigurations, unpatched systems), Malicious Users or Spies, Abuse of Privileges, Data Leakage, Spear Phishing, etc.

are being resolved by technical solutions or secure frameworks (El-Mhamdi et al., 2022). Besides, more and more users start to worry about their personal and sensitive information being leaked by LLMs. Therefore, some actions have been taken to cope with that and ensure output is legal (Raeini, 2023). Finally, it is important to acknowledge that GPT-3-like models typically require high-performance computing environments due to their extensive model sizes and computational demands, potentially hindering their deployment on local systems at a low cost. In contrast, BERT-like models are more compact and well-suited for use in resource-constrained settings. Nonetheless, recent advancements in gradient optimization techniques and model simplification strategies have been proposed to address this issue (Frantar and Alistarh, 2023), rendering LLMs feasible for adoption by micro-enterprises and medium-sized enterprises.

2.2. Cyber threat detection

Since its emergence in the 1960s, the Internet has continually faced a myriad of threats such as viruses, vulnerabilities, and Trojans. Simultaneously, the evolving nature of cyberspace encompassing digital technologies such as computers, communication networks, Internet of Things (IoT), etc., has correspondingly made cyberspace threats more complex and unpredictable. Generally, cyberspace threats (Aslan et al., 2023) refer to all events that have negative impacts on cyberspace security, ranging from natural disasters to management oversights and hackers' activities. The origins of threats include individual hackers, criminal organizations, and even other countries. Nearly all threats achieve it by exploiting technical vulnerabilities, system weaknesses, or human errors. According to their source, cyberspace threats can be divided into internal threats and external threats, with typical examples shown in Table 2.

To address these threats, abundant threat detection and discovery works have been developed. The process of cyberspace threat detection (Al-Mhiqani et al., 2020) is comprehensive and changing, normally including the identification, evaluation, and prediction of various potential or known malicious entities or activities within cyberspace. To learn the current landscape in this field, we have systematically reviewed some representative-related works as follows.

In the early IDS or threats detection tasks, security researchers predominantly utilized Misuse Detection skills such as signature-based and rule-based methods, while also employing a limited use of Anomaly Detection skills like heuristic and statistics-based methods. On one hand, signature-based methods, relying on known threat characteristics, effectively identify threats by matching attributes of entities or behaviors against databases. The Rule-based detection achieves identifying potential threats by predefined rules, indicating attacks when network traffic

exhibits certain abnormal patterns (Kumar et al., 2020). Conversely, heuristic analysis is more flexible, focusing on threat patterns rather than fixed characteristics, thus protecting against new and unknown threats. The statistics-based methods emphasize interpreting system state changes in networks and learning expected system behaviors to assess threat-containing traffic anomalies (Cao et al., 2017). Despite their efficacy in single-task detection scenarios, these methods are often trapped with high false positive rates (FPR), limited adaptability, dependence on known threats database, and challenges in effectively countering unknown threats.

With the advancement of AI applications, machine learning (ML) has driven the development of threat detection. ML for threat detection is primarily divided into supervised and unsupervised learning. On one hand, supervised learning focuses on classifying entities into threats or non-threats based on certain feature sets, such as differences between secure and threat data, and similarities among various threat data. There are plenty of examples of using ML in threat detection like Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Naive Bayes Classifiers (NB), and Logistic Regression (LR) (Almaleh et al., 2023). On the other hand, unsupervised learning emphasizes classifying objects into different categories based on structural features within the data, with related works like K-means, DBSCAN, Hierarchical Clustering, and Gaussian Mixture Models (Hou et al., 2022) for threat clustering. Although unsupervised learning does not rely on labeled datasets, reducing data dependency and human error, its outcomes require further analysis and are sensitive to noise and outliers. It is shown that ML is effective in analyzing superficial features, mitigating some inherent issues in early methods. However, ML may be inadequate when encountering large, complex datasets with in-depth threats. Moreover, ML usually requires extensive labeled data or domain expertise for training, posing practical application challenges in detecting new threats.

Owing to the complexities of threats characterized by non-linearity and high dimensionality, DL has been integrated into threat detection, leveraging different kinds of NNs and their variants to accommodate intricate feature engineering and tasks. The optimized Convolutional Neural Network (CNN) and Recurrent Neural Networks (RNN) have demonstrated efficacy in mining spatial and temporal features of threat (Ahmed Issa and Albayrak, 2023), thus enabling comprehensive network traffic analysis and continuous monitoring for APT. Besides, the Transformer, acclaimed for its proficiency in processing large volumes of sequential data, is well-suited for extracting threat content from diverse threat carriers (Xing et al., 2023). Likewise, the Graph Convolutional Network (GCN) has shown potential in analyzing the network's structure and nodes' relationships, especially in threat detection by provenance graphs (Li et al., 2023b). Besides, Some variants of NNs like Generative Adversarial Networks (GANs) and Deep AutoEncoders (DAEs) have been proven to be efficient in the real-time discovery of new threats. Despite the unparalleled advantages of DL in identifying APT and dynamic threat analysis, there are still challenges, including high training and computing costs, limited interpretability of the model, and optimization problems, most of which remain largely unresolved to date.

Concurrently, some evolving technological resolutions are also put forward in this field. Firstly, Reinforcement Learning (RL), an AI paradigm predicated on learning and optimizing decisions through environmental feedback, plays a pivotal role in large-scale integrated threat detection systems (Yu, 2022). Through refined reward functions, RL incessantly adjusts and optimizes detection strategies, thereby enhancing the online detection efficacy. In conjunction with other algorithms, RL contributes to the autonomous identification and mitigation of threats, facilitating threat hunting and adapting to new attack patterns significantly. In addition, visualization technologies and tools demonstrate extensive utility in analyzing large-scale events, traffic, logs, etc. Upon the visual models, computer vision (CV) algorithms can be used to analyze and identify threat entities, aiding security analysts in more intuitively learning and majoring security data and

threat patterns (Nikolov and Mees, 2023). Moreover, the development of threat detection based on Cyberspace Threat Intelligence (CTI) has reached new levels of maturity. From real-time and historical threat data, CTI can adeptly encapsulate advanced knowledge of diverse threats, aiding in cyberspace situational awareness. CTI can automate the identification of known threats and be used in the inference and prediction of unknown threats (Jiang et al., 2023). In all, These methodologies have significantly complemented previous methods, markedly curtailing the duration to discover attacks and amplifying the adaptability of threat detection systems. Nevertheless, each of them still possesses challenges needing further work, such as the complexity of reward functions designed in RL, the effectiveness of visualization tools on massive data, and the demands for data quality and timeliness in CTI, etc.

Last but not least, an array of cutting-edge technologies are increasingly coming to the fore. Initially, threat detection based on cloud computing and big data is being more widely implemented within cloud security products (Habeeb et al., 2019). Big data skills, such as data lakes and distributed computing, are employed for the efficient analysis of massive volumes of security data like logs, aiding security researchers in swiftly pinpointing and addressing threats. Subsequently, integrated methods amalgamating different threat detection approaches are emerging as a new trend in strategic solutions (Akram et al., 2023). These methods combine ML, DL, behavioral analysis, etc., to enhance the accuracy, comprehensiveness, and adaptability to real-world network environments, thereby reducing FPR and FNR. Apart from these, threat modeling and prediction are also providing robust support for the discovery of unknown threats (Zhao et al., 2022). Modeling in-depth information and analysis of attackers' behaviors and vulnerabilities, not only assist researchers in identifying real-time threats but also in forecasting potential threats and vulnerabilities effectively. Collectively, this avant-garde research is providing new resolutions for threat detection and discovery, promoting trends of threat detection towards high-level automation, high generalization, and cloud integration. However, these technologies are still in their nascent stages, facing various application challenges necessitating ongoing optimization.

Up to now, multiple threat detection and discovery methods have been introduced. Although there are still a few other types of these not mentioned, the studies summarized above are sufficient for us to learn research status in this field. It is shown that previous methods confront different challenges to some extent, especially in complex scenarios like cloud security. For instance, traditional threat detection and discovery methods often struggle with high FPR or FNR, indicating difficulties in discovering unknown threats, adapting to evolving threats, being misunderstood by constrained context, etc. These limitations hinder their effectiveness in the dynamic and data-rich landscape of cyberspace. Luckily, most of such problems are being solved by LLMs, renowned for their context-aware processing, comprehensive knowledge bases, and adaptive learning ability. Recent studies suggest that LLM could improve the accuracy, adaptability, and generalization of threat detection in dynamic and data-intensive environments.

3. Thematic works

This Section will delve into multifaceted research utilizing LLMs for threat detection in specific scenarios. We did not categorize thematic works from another perspective (e.g. input data types, different stages) because we aim to identify security scenarios most closely related to LLMs, facilitating rapid comprehension of technological dynamics in various research fields. Meanwhile, we found limited literature indicating LLMs are directly applied to entire stages of threat detection. Conversely, most studies have focused on utilizing LLMs to optimize specific processes within established research frameworks. In light of this, the role of LLMs in various security scenarios may be more

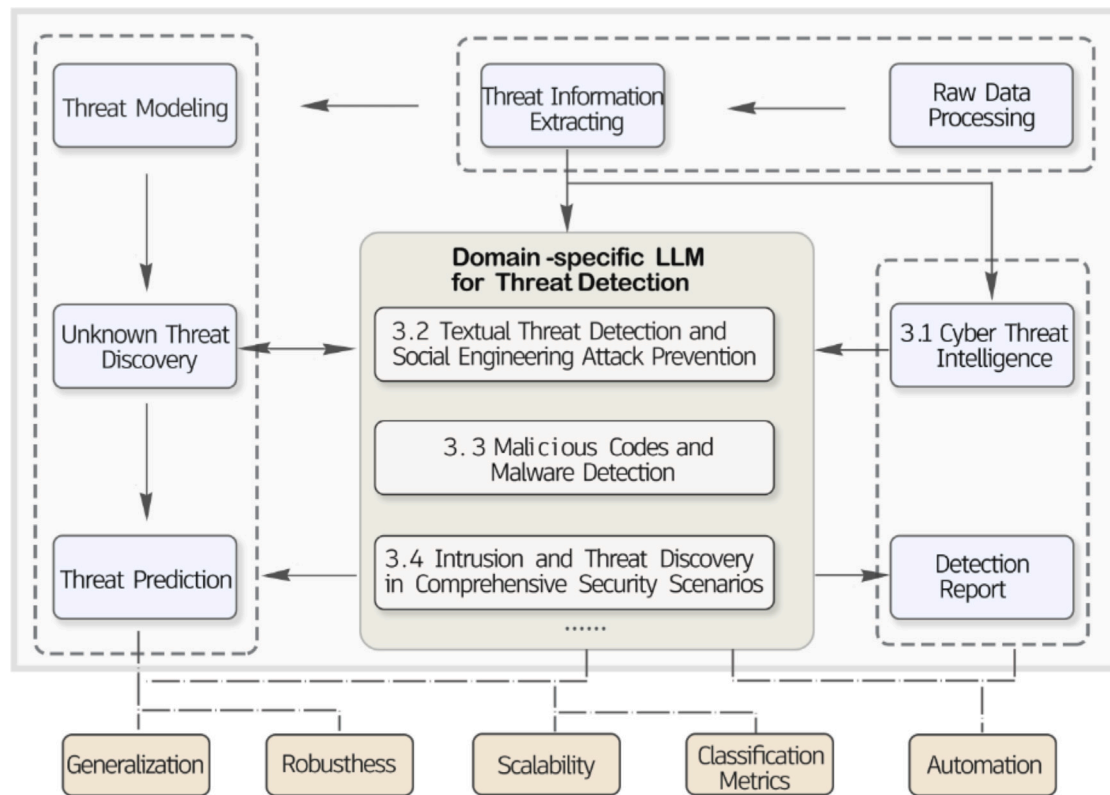


Fig. 1. Framework of thematic works.

important than the problems LLMs process. The organization of this section will adhere to the guidelines outlined in Fig. 1.

As depicted in Fig. 1, thematic works have been categorized into four security scenarios that are intricately linked to the application of LLMs. These security scenarios encompass correlative works of CTI (3.1), textual threat detection and social engineering attack prevention (3.2), malicious codes and malware detection (3.3), and intrusion and threat discovery in comprehensive security scenarios (3.4). Simultaneously, leveraging LLMs in CTI production enables CTI to facilitate diverse cyber threat detection processes under the supervision of LLMs. Additionally, LLMs can be utilized for generating detection reports and other security-related documentation. According to Goodfellow et al. (2016), the development of AI systems typically involves sequential phases such as data collection, data processing, model construction and training, model evaluation, model deployment, and feedback improvement. This framework is also applicable to the implementation of LLMs in various security contexts, where the process can be delineated into stages encompassing data collection and processing, extraction and analysis of threat features, and execution of target tasks such as classification or clustering for threat discovery, regression for threat prediction. So we can discuss which role LLMs played in different security scenarios step by step. Moreover, some evaluation criteria such as classification metrics (e.g. accuracy, precision, recall, F1 score), scalability, and robustness were used to estimate the performance improvements for previous threat detection models by LLMs. In this way, can this review effectively help researchers use LLMs for cyber threat detection in a targeted manner.

3.1. Applications of LLMs on CTI

To start with, we know LLMs were initially tailored for NLP tasks, enabling their utilization in tasks such as name entity recognition (NER) and relationship extraction (RE) on unstructured data, thereby facilitating the generation of structured textual content. Consequently,

numerous researchers have already employed LLMs in the development and implementation of CTI.

According to the definition by Gartner, CTI, intelligence strongly related to threat detection, refers to all evidence-based knowledge including mechanisms, labeling, actionable recommendations, meaning, context, etc., which is relevant to existing threats or potential threats faced by assets. Therefore, CTI has a high value on threat detection and response. In a narrow sense, CTI mainly includes trap identification used to identify and detect threats, such as file HASH, IP, domain name, program running path, registry key, and related attribution labels. As outlined by Nova (2022), The life cycle of CTI can be divided into six stages: CTI targets, CTI collection, CTI processing, CTI analysis, CTI sharing, and CTI feedback. We found LLMs have predominantly been utilized in stages of CTI collection, processing, and analysis up to now, as shown in Fig. 2. In the remaining subsection of thematic works, we also use highlighted program blocks to indicate which part LLMs are involved in.

A. LLMs for CTI Collection

In most cases, the challenge of identifying unknown threats is often exacerbated by the scarcity of high-quality data for CTI and the limited generalization capabilities of detection methods. To effectively collect CTI that aligns with the specific requirements of a network, it is imperative to have access to both sufficient quantities and high-quality labeled data. Bayer et al. (2023) fine-tuned BERT and utilized data augmentation with text-davinci-002, combining three strategies of transfer learning, data augmentation, and few-shot learning to train high-quality classifiers from a few labeled instances. In the data augmentation phase, GPT-3 generated new instances from a few existing labeled examples, significantly reducing the data labeling workload. Results have shown that with a limited number of data instances (testing with 32 instances), the classification results were comparable to those trained with 1800 instances, highlighting its effectiveness in assisting data labeling in CTI collection and threat detection. Additionally, the study noted that a large amount of CTI data in open-source

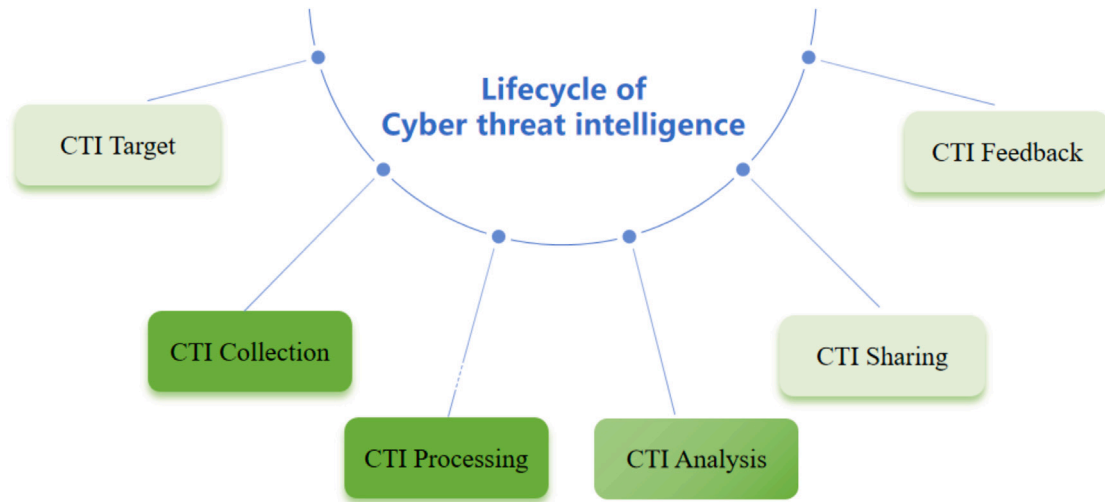


Fig. 2. A diagram about LLMs in Three Key CTI Lifecycle Stages.

databases mostly comprises Indicators of Compromise (IoCs) like file hashes and IP addresses. Since this information is only short-term effective and can be easily modified by hackers, most researchers prefer focusing on capturing data applicable to threat analysis rather than collecting all IoCs, which may be assisted by LLMs.

Jo et al. (2022) constructed the Vulcan system, which was designed to extract various types of raw data describing CTI from unstructured text and determine the starting semantic relationships. The methodology employed an optimized BERT model for RE and NER tasks related to various threats. Following the training of the models, six types of Cyber Threat Intelligence (CTI) data and seven relationships among them were successfully identified. Testing on a self-constructed dataset containing 6,791 entities and 4,323 relations revealed that Vulcan achieved an F1-score of 0.972 in NER tasks and 0.985 in RE tasks, facilitating rapid analysis of cyber threat entities and their associations. This high level of accuracy significantly reduced the time and cost burden on security researchers. Furthermore, the study highlighted the importance of knowledge graphs in enhancing the representation of knowledge, which is commonly used in CTI collection and processing.

Ren et al. (2023) conducted a cyberspace security platform named CSKG4APT based on knowledge graphs. For unstructured text, CSKG4APT used a CTI View based on BERT, constructing APT knowledge graph models by collecting and analyzing OSCTI and building fragmented information into APT scenarios containing 12 node types and 7 logical relationships. For updating the knowledge graph, an APT threat knowledge extraction algorithm was proposed, utilizing DL and expert knowledge to distinguish data associated with specific threats from irrelevant content. The framework provided attribution and countermeasures for APTs, demonstrating that CSKG4APT is proactive in security confrontation. It can collect and integrate massive fragmented intelligence using LLMs to update security strategies. Additionally, the study highlighted two main ways of presenting CTI: IoCs and unstructured reports.

Zhou et al. (2022) optimized CTI View, an automated system for extracting and analyzing CTI text published by various security vendors. To achieve this, a CTI extraction framework was designed based on automated testing, text recognition, and denoising. This framework significantly improved the adaptability of crawlers to crawl heterogeneous CTI. For threat entity extraction, a composite model named BERT-BiLSTM-GRU-CRF was used, achieving better results on the labeled dataset compared to other existing models. Additionally, the system effectively extracted IoC and TTP information described in CTI using regular expressions combined with a black-and-white list mechanism.

Drawing from the literature reviewed above, it is hypothesized that LLMs have the potential to effectively process security-related content such as news articles and blogs from various origins, validate the accuracy of the information, identify crucial threat indicators, and subsequently incorporate them into comprehensive analyses of attack events or other forms of intelligence data.

B. LLMs for CTI Processing and Analysis

To process the raw CTI data obtained, further steps are required. Wadhwa et al. (2023) evaluated the performance of LLMs in RE tasks by using GPT-3, Flan-T5, and other LLMs not previously explored in existing works. They tested these models on established RE datasets such as ADE, CoNLL, NYT, and DocRED, with varying degrees of supervision. The methodology involved treating RE as a sequence-to-sequence assignment, representing relations between entities as target strings, and incorporating human evaluations to tackle the complexities associated with assessing generative models for RE. They found that few-shot prompting with GPT-3 achieved near state-of-the-art (SOTA) performance. Additionally, despite its initial inferiority, a smaller model like Flan-T5 also performed well when fine-tuned using Chains of Thought (CoT) explanations generated by GPT-3. It demonstrated LLMs can extract hidden and knowledge-level relationships from extensive security texts.

Sewak et al. (2023) developed CRUSH, utilizing GPT-4 and a semantic hypernetwork (LLM-EKG) to design and enrich Threat Intelligence Graphs (TIGs) for improved CTI processing and threat detection efficacy. This model effectively automated the extraction, inference, and validation of threat information from various sources, particularly excelling in identifying malicious scripts and mapping threat techniques. The CRUSH method achieved up to a 99% recall rate in detecting malicious scripts and maintained over 90% recall in identifying specific threat types. Additionally, CRUSH could standardize threat behaviors, enhancing the accuracy and efficiency of threat detection. The study also emphasized the importance of vulnerabilities in CTI, highlighting how comprehensive vulnerability analysis provides critical information for threat detection, including exploitable vulnerabilities, potential threats, attacker techniques, and targets.

To process CTI of vulnerability better, Liu et al. (2023) used ChatGPT to map vulnerabilities to CWE-ID and ATT&CK technical IDs through a process termed Vulnerability Description Mapping (VDM). The results showed that ChatGPT performed better in mapping vulnerability descriptions to CWE-ID, while it struggled with mapping to ATT&CK technology IDs. A significant issue was identified in ChatGPT's

Table 3
Thematic works about LLMs on CTI.

Tasks	Effectiveness	Limitations	literatures
CTI Collection	improving data labeling and data enhancement by GPT3.	Data enhancement and categorization worked badly when the labels were too similar.	Bayer et al. (2023)
	High F1-Scores for NER and RE tasks related to various threats.	Failure to validate the reliability of data and novel data beyond training.	Jo et al. (2022)
	Effective integration of fragmented intelligence.	Inability to online knowledge completion and construction of knowledge, the semantic gap between some IoCs.	Ren et al. (2023)
	Using BERT to extract entities from heterogeneous CTI and update them into APT Threat Knowledge Graphs.	Inability to accurately locate entity boundaries, incomplete extraction of entities.	Zhou et al. (2022)
CTI Processing and Analysis	few-shot prompting with GPT-3 or smaller models prompted by CoT of GPT-3 achieves SOTA performance.	Inability to perform severely on complex datasets; GPT3 without fine-tuned can be accessible only via interface.	Wadhwa et al. (2023)
	LLM-EKG achieves high recall on the task of detecting malicious scripts.	length for text analysis is limited.	Sewak et al. (2023)
	Vulnerability mapping to CWE works better than mapping to ATT & CK tech IDs.	ChatGPT cannot be used directly for VDM, but 'closed source is not the end of story'.	Liu et al. (2023)
CTI Generation	Automated report generation with high recall rate.	Relying on ChatGPT may cause cost and privacy concerns.	Perrina et al. (2023)
	Automated CTI extraction to generate structured STIX reports.	LLMs' hallucination problem in some tasks.	Siracusano et al. (2023)
	Automated Report Generation.	hallucination problem in some tasks.	Gupta et al. (2023)
	Automatically generate high-quality CTI to meet the organization's needs.	Not mentioned	Mitra et al. (2024)

conceptual understanding of CAPEC and ATT&CK technologies. It was hypothesized that the high diversity of ATT&CK technologies and the poor quality of the data contributed to GPT's inability to understand the relevant technical content properly, resulting in a high rate of false positives. Consequently, the study concluded that GPT might not be the best option for VDM tasks.

C. LLMs for CTI Generation

It is always labor-intensive work to manually produce security reports other than CTI. However, LLMs are competent in language analysis and generation capabilities, making them suitable for generating structured CTI. To make it, Perrina et al. (2023) proposed AGIR, an automated intelligence report generation tool using ChatGPT to generate STIX-standard CTI. This tool achieved a 99% recall rate without introducing LLM hallucinations. By combining this information extraction system with AGIR, the study demonstrated that it can automatically process multiple reports, significantly facilitating the automation of CTI tasks.

To generate CTI from a wider range of sources, Siracusano et al. (2023) introduced a new large-scale open benchmark data set and aCTIon, a structured CTI information extraction tool. This data set comprises 204 real-world public reports and their corresponding structured CTI information in STIX format. Researchers utilized GPT-3.5 to design prompts for two custom information extraction tasks: NER and attack pattern recognition. They conducted self-checks based on the output results and validated these results to ensure accuracy. Once verified, security reports were produced in standardized STIX format. The study demonstrated the effectiveness of aCTIon in automating the extraction of CTI from public reports, providing a robust benchmark data set for future research and applications. In the same light, Gupta et al. (2023) also argued that ChatGPT can automatically generate CTI based on a variety of data sources, including social media, news articles, dark web forums, and other online sources. By processing and analyzing large amounts of data, ChatGPT can generate reports that are accurate, comprehensive, and easy to understand. These reports help organizations identify potential security threats, assess their risk level, and take appropriate measures to mitigate them.

Mitra et al. (2024) proposed LOCALINTEL, a system that retrieves CTI from global threat repositories based on prompts and contextualizes it to a specific organization using its local knowledge database.

LOCALINTEL operates through three key phases: global CTI retrieval, local knowledge retrieval, and contextualization completion generation. The global CTI retrieval phase gathers intelligence from the global threat repository, while the local knowledge retrieval phase extracts relevant information from the local knowledge database. Finally, the fusion of these knowledge sources is orchestrated through a generator to produce contextualized completion results. This system automates the accurate customization of global threat intelligence against local data, helping Security Operations Center (SOC) analysts reduce manual workload. It has shown that LLMs can effectively handle massive different knowledge sources, processing global and local knowledge databases to automatically generate organization-specific CTI.

Except for CTI, some materials indicated that LLMs can also be utilized in generating threat detection reports, cyber security posture reports, and other security documentation, all of which serve as aids for threat detection indirectly. From this part, it is evident that LLMs hold significant promise in enhancing the automation of CTI production and the generalization capability of CTI aggregation, thereby offering substantial informational support for threat detection-oriented CTI. All related works mentioned in this part are summarized in Table 3.

3.2. Textual threat detection and social engineering prevention

It is well-established that the primary input data utilized by LLMs consists of natural language. Consequently, threat detection in textual content represents one of the most pertinent threat detection scenarios for LLMs. Social engineering attacks, which commonly exploit communication mediums reliant on text content such as messages, emails, and public statements, serve as prime examples. As a result, some researchers have leveraged LLMs for the detection of social engineering attacks, particularly phishing attacks, yielding favorable outcomes.

As illustrated in Fig. 3, LLMs are primarily utilized for text collection and detection purposes. In the text collection phase, a segment of the training data for the detector is sourced from open-source community media and text datasets, some of which are processed by LLMs. However, another segment may be generated or created by LLMs independently. In the phase of model training and detection, certain researchers may decide to fine-tune existing LLMs, while others may

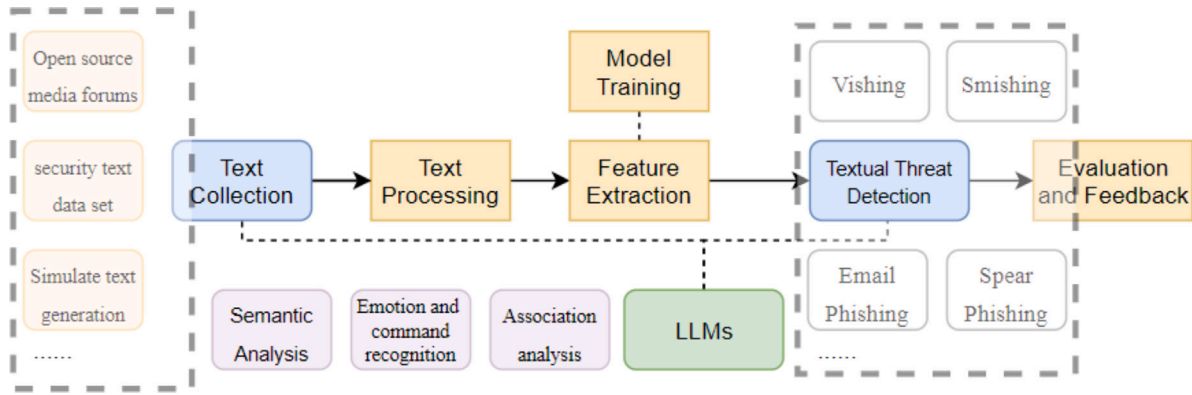


Fig. 3. A diagram about LLMs' application in textual threat detection.

opt to retrain classifiers and extract threat features using their datasets, especially in the context of BERT-style models.

A.LLMs for Threat Text Collection

In terms of data imbalance, LLM can generate extra raw data for training LLM or non-LLM detection models, thereby improving the overall generalization ability of threat detection. To expand the collection of threat text from a broader range of sources, Jin et al. (2023) introduced Dark BERT, a highly efficient and scalable LLM specifically trained on a vast of text sourced from the dark web. This model was designed to effectively identify emerging threats within the darknet. Dark BERT demonstrated exceptional performance in the areas of monitoring, investigating, and tracking illicit activities, thereby showcasing its potential to enhance threat intelligence derived from the dark web. Besides, it can raise public awareness regarding the dark web and the prevalence of social engineering attacks. However, the study acknowledged that not all threat texts are suitable for Dark BERT. It currently faces challenges with bias, cost, and privacy protection, which have not been fully resolved.

Sharma et al. (2023) experimented to study whether phishing emails generated by humans and GPT can be identified, and to determine the differences in attack effectiveness. Through the implementation of over 40 practical trials involving distinct cohorts, the researchers observed a marginal discrepancy in the success rates between manually composed emails and those generated by the GPT-3 system. However, the results may be restricted by the content, type, and implications caused by the experimental design of these emails. Future research could explore the potential utility of LLMs for the generation of more sophisticated adversarial samples to enhance detection capabilities.

Simultaneously, Tsinganos et al. (2023) proposed SG-CSE BERT and a framework to identify the real intention of the interlocutor, which is essential for detecting social engineering attacks based on dialogue. The framework leverages LLMs to generate and analyze historical information, allowing them to output high-quality predicted attack states. This approach enhances the discovery of social engineering attacks by supplying more hidden information to focus on the underlying intentions in conversations.

Shibli et al. (2024) presented a technical framework of phishing attacks based on prompt injection attacks using AbuseGPT and existing conversational models like Bard and ChatGPT. This study demonstrated the creation of smishing texts and real-world crafter smishing campaigns, providing a vivid example of how LLMs can be used to generate threat text effectively. This framework highlights the potential misuse of LLMs in generating sophisticated phishing attacks, emphasizing the need for robust security measures.

Notably, Security Tools like FraudGPT and WormGPT can also generate abundant threat content that is not limited to malicious text. These weaponized generative AI tools can effectively assist hackers,

such as malicious code/malware generation, phishing website creation, vulnerability analysis, etc. Since they are anonymous, illegal, and hard to find, few researchers chose to talk about them directly. But that does not mean we can ignore these tools because they may enable more attackers with the ability to carry out APT. Meanwhile, they have enormous application value in enriching threat corpora, indirectly enhancing threat detection capabilities.

B.LLMs for Threat Text Detection

In this section, some applications of LLMs or their APIs on threat detection for textual threat detection will be introduced. Ghourabi and Alohaly (2023) combined Transformer and Ensemble learning with GPT-3 to detect spam and phishing messages, achieving a classification accuracy of 99.91%. This high level of accuracy demonstrates the potential of integrating advanced machine learning techniques with LLMs for effective threat detection. However, the study also mentioned some limitations, such as the detection objects being primarily English messages, the poor scalability of models, and the small amount of test data used in the experiments.

To investigate the adversarial capabilities of LLMs for phishing attacks, Roy et al. (2023) conducted an evaluation of ChatGPT, Google Bard, and Claude in their capacity to generate phishing content. The researchers employed a series of malicious prompts to assess the models' proficiency in creating phishing websites and emails. The methodology encompassed the creation of prompts that directed the LLMs to generate both conventional and evasive phishing attacks, followed by an evaluation of the produced content using qualitative and quantitative measures. They also developed a BERT-based detector to identify and prevent the generation of malicious prompts in real time. The study found that LLMs can generate highly convincing phishing content without prior adversarial modifications. The detection tool achieved an accuracy of 96% for phishing website prompts and 94% for phishing email prompts, highlighting the enormous potential of LLMs in phishing attacks and defense.

Nahmias et al. (2024) focused on improving feature representation for phishing emails and designed a problem list to enable LLMs to infer different attributes of the emails. Based on these inferences, a feature representation was constructed for use in a downstream machine-learning classifier. The test results indicated that the accuracy of the detection exceeds 87% and the F1-score exceeds 91%. This method demonstrates good generalization ability and can be further optimized through model tuning or case analysis of misclassified samples.

Kumarasiri et al. (2023) proposed Cybersmish, a system utilizing MobileBERT, DistilBERT, and TF-BERT to enhance the detection of smishing attacks on mobile devices. Cybersmish consists of three main components: a message content analyzer, a URL analyzer, and an HTML content analyzer. By leveraging these pre-trained models, Cybersmish achieves over 90% accuracy in analyzing SMS content, classifying

Table 4
Thematic works about LLMs on Textual Threat Detection and Social Engineering Prevention.

Tasks	Effectiveness	Limitations	Literatures
Threat Text Collection	Training DarkBERT on dark web ;Performed well in text generation.	Limited in non-English tasks; There is a strong dependence on domain data.	Jin et al. (2023)
	Phishing emails by GPT-3 are slightly inferior to manually crafted ones.	Results may be limited to the content, type, and other factors in the experiment.	Sharma et al. (2023)
	SG-CSE BERT for analyzing historical information and predicting attacks.	Despite the of convenience deploying, it has a small scale and average accuracy.	Tsinganos et al. (2023)
	A framework for phishing attacks using existing conversational models and an instantiation testing process is provided.	Influenced seriously by prompt injection; the quality evaluation is not complete; Method has short-term effectiveness.	Shibli et al. (2024)
Threat Text Detection	By combining Transformer and Ensemble learning with GPT-3, the accuracy of email classification became higher.	Data set only involves English messages; the scalability of the detection model is poor.	Ghourabi and Alohalay (2023)
	Using different types of LLMs to generate and detect phishing content.	Relying on prompt; Poor real-time performance; Restricted samples.	Roy et al. (2023)
	Designing problems, LLMs can infer attributes, and construct feature representations for downstream classifiers.	Not Mentioned	Nahmias et al. (2024)
	Effective in message content analysis, URL classification, etc.; Accurately identify phishing for mobile devices.	Not Mentioned	Kumarasiri et al. (2023)
	Using basic models such as GPT-4, the accuracy of phishing website detection tasks reached a high level.	Results are unstable and susceptible to prompt or injection attack; The training data has a deadline time node.	Koide et al. (2023)
	The accuracy, recall, and F1-score of threat and hate speech detection on social media are superior to traditional and DL models.	Language types for detecting have certain limitations; when applied to other languages, the model needs to be retrained.	Bilal et al. (2023)

URLs, and evaluating HTML authenticity. This work highlights the effectiveness of advanced machine learning and LLMs in protecting users from social engineering attacks and enhancing mobile device security.

Koide et al. (2023) conducted a system named Chat Phish Detector, which started with collected seed URLs to crawl for phishing and other websites, directly using ChatGPT to detect and achieve an accuracy rate of 98%. This research demonstrated that LLMs performed well in detecting various social engineering threats. However, the study also noted that detection results are unstable and susceptible to prompt injection attacks, and the training data has a deadline. Despite these challenges, the accuracy, especially for GPT-4, was found to be good.

Bilal et al. (2023) targeted hate speech issues on social media, developing a threat speech classification system based on BERT and Roman Urdu. Trained on a large dataset of 173,714 texts in this unique language, the study compared traditional machine learning and deep learning models, including LSTM, BiLSTM, BiLSTM with attention layer, and CNN. It also explored transfer learning by combining pre-trained BERT embeddings. The results showed that the transformer-based model outperformed traditional and deep learning models in terms of accuracy (96.70%), precision (97.25%), recall (96.74%), and F1-score (97.89%). This demonstrated the model's stronger generalizability across cross-domain datasets. LLMs possess a robust semantic comprehension and a capacity for human emotional understanding, enabling them to potentially detect misinformation, harmful content, and biased information. Maybe this capability is essential in safeguarding information content security against potential cyber threats.

Furthermore, some other reports may declare LLMs can be utilized for internal threat detection, monitoring social forum speech, conducting sentiment analysis of threats, and recognizing intentions. As advanced models, LLMs possess capabilities such as profound semantic comprehension, human-like reasoning, and sentiment analysis, enabling them to identify and analyze attackers' motives and behavioral patterns more effectively. This comprehensive approach enhances the accuracy and generalization of textual threat detection. Meanwhile, some limiting issues in existing works are being further studied these days. All related works in this part are summarized in Table 4.

3.3. Malicious code and malware detection

Code fragments in programs can be regarded as special strings or sequences that are restricted by syntax rules and internal logic. Quite a few studies or security products (e.g. Copilot, Deep Code) on code analysis have indicated that DL models can effectively extract hidden information from codes to be used for vulnerability mining or security issues analysis. In this view, some researchers used LLMs to detect and analyze malicious codes or malware. It should be noted that there is a certain difference between detection works and vulnerability mining. Vulnerability mining focuses on discovering and exploiting flaws or bugs in system programs, but threat detection for codes focuses on analyzing whether programs are harmful. Although the detection of malicious codes and malware is highly correlated with comprehensive security scenarios, we still chose to discuss this topic separately. Because abundant literature has shown that using LLMs for security analysis on codes is as appropriate as semantic analysis on natural language.

As depicted in Fig. 4, LLMs are primarily used for the acquisition of software codes and data, as well as for static detection of malicious codes or malware. LLMs can process code fragments sourced from open data repositories, and network crawlers, and address imbalanced training datasets. Additionally, LLMs have the potential to simulate malicious activities, generate malicious codes or instructions, and even create simple malware or other tools. Subsequently, LLMs can be employed for the static detection of these malicious entities, which involves identifying potential threats through the examination of software files, code, and structure.

A.LLMs for Malicious Code and Malware Collection

It is highly effective in some adversarial tasks by generating supplementary datasets containing specific features. LLMs are competent for generating high-quality code and tools, which can enhance the comprehensive detection capabilities of various models for malicious code or malware detection. Botacin (2023) investigated the capacity of ChatGPT to produce malicious software directly utilizing GPT-3. Their findings indicate that although GPT-3 encounters challenges in generating comprehensive malware samples based on detailed descriptions, it can proficiently assemble malware by utilizing modular descriptions and generate numerous variations of malicious software. This research

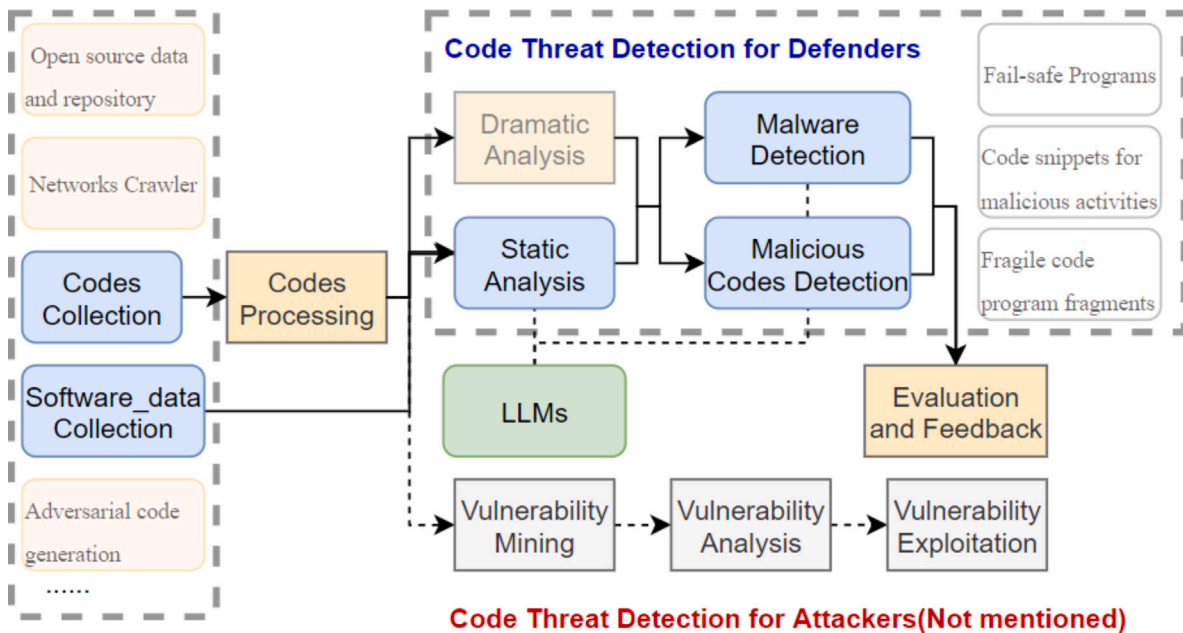


Fig. 4. A diagram about LLMs in Malicious code and Malware detection.

underscores the potential for GPT-3 to be misused in the creation of a wide range of malware, suggesting that generative LLMs could serve as potential generators in certain adversarial detection frameworks.

To illustrate examples of specific security vulnerabilities, [Charan et al. \(2023\)](#) systematically produced executable code snippets for the top 10 MITRE techniques of 2022 utilizing ChatGPT. Through a comparative analysis of its efficacy in generating malicious scripts with Bard, they revealed that GPT exhibited a higher susceptibility to misuse.

[Pa Pa et al. \(2023\)](#) used ChatGPT and Auto-GPT to generate seven malware programs and two attack tools, demonstrating the possibility of creating functional malware within 90 min, including debugging time. Through prompt engineering, ChatGPT was able to create executable malicious codes, showing greater responsiveness and debugging capability. Concurrently, a series of adversarial detection frameworks based on LLMs are gaining popularity in the fields of automated offense and defense, as well as adaptive threat detection.

[Hu et al. \(2021\)](#) explored deep learning-based malware detection and proposed a novel Adversarial Malware Generation (AMG) framework named MalGPT. They used GPT to analyze malware byte sequences, achieving successful single-query black-box evasion models, and effectively circumventing malware detectors with just one query. MalGPT exhibited outstanding performance in handling eight major malware categories, with an evasion rate exceeding 24.51%, significantly outperforming existing benchmark methods. This study also incorporated malware source code into perspective analysis, suggesting that this multi-view deep learning approach could further enhance evasion capabilities.

[Devadiga et al. \(2023\)](#) proposed the GLEAM model, which combined hex code and opcode features with LLM embeddings to generate synthetic samples similar to evasive malware. By using LLMs to create coherent and structured samples, they increased the average evasion rate by 25.0%, demonstrating the ability of LLMs to generate adversarial malware samples. These techniques not only aid in generating security test cases and swiftly uncovering new threats, but they also hold significant value for understanding and combating malware, especially in real-world detection scenarios with data imbalances.

B.LLMs for Malicious Code and Malware Detection

In addition to applying LLMs to the generation stage of malicious samples, some researchers opt to utilize LLMs in the static analysis

or direct detection stage of malicious codes or malware on rare occasions. [Demirci et al. \(2022\)](#) used a combination model of Stacked BiLSTM, DSLM-GPT2, and GLM-GPT2 for malicious code and malware detection. This combined approach proved significantly superior to other deep learning models, particularly in detecting malicious behavior in PE files. The study highlighted the effectiveness of integrating multiple advanced models to enhance the accuracy and reliability of malware detection systems.

[Gao et al. \(2022\)](#) conducted an analysis in which Control Flow Graphs (CFG) were extracted from Portable Executable (PE) files and utilized the large-scale pre-trained model MiniLM to derive node features of the CFG. Subsequently, these features were converted into feature vectors and subjected to classification through a multi-layer perceptron. The study showcased the efficacy of integrating CFG extraction with sophisticated pre-trained models such as LLMs to enhance the accuracy of PE file analysis.

[Li and Gong \(2023\)](#) employed GPT-3.5 and ChatGPT in the context of identifying malicious web pages through the analysis of accessed source codes. Their empirical findings demonstrate that the efficacy of these LLMs is on par with, if not superior to, conventional deep learning approaches, thus affirming the viability of leveraging LLMs for the detection of network security risks. This investigation underscores the promise of LLMs in bolstering cybersecurity protocols by facilitating the detection of potential threats.

[Li et al. \(2023a\)](#) employed the Llama-7b model and image analysis of PE files for ransomware detection. By converting PE files into grayscale bitmap images, they analyzed these images using the Llama-7b model. This method achieved a significant true positive rate and minimal false positives and negatives, proving effective for detecting ransomware variants. The study demonstrated the potential of using image analysis in combination with LLMs to enhance ransomware detection.

[García-Soto et al. \(2022\)](#) detected malicious behavior in Android applications using the CodeT5 pre-trained language model. They first generated embeddings from chunks of the source code after decompiling the Android application to Java, then used these embeddings for the BiLSTM classification model. This approach effectively combined LLMs with existing malicious code and malware detection methods, demonstrating the feasibility and potential of using LLMs in enhancing Android malware detection.

Table 5
Thematic works about LLMs on Malicious Codes and Malware Detection.

Tasks	Effectiveness	Limitations	literatures
Malicious Codes and Malware Collection	GPT-3 can construct malware via building block descriptions and generate multiple malware variants.	GPT-3 has trouble generating entire malware samples from complete descriptions	Botacin (2023)
	ChatGPT can accelerate the generating of malicious code.	Reluctance in generating malicious code for corresponding.	Charan et al. (2023)
	ChatGPT generates functional malware (up to 400 lines) within 90 min.	AugoGPT does not ease the hurdle of generating the right prompts for malware.	Pa Pa et al. (2023)
	Significantly improves the chance of single-shot evasion.	Excessive reliance on querying a malware detector multiple times.	Hu et al. (2021)
Malicious Codes and Malware Detection	Effectively expanding the space for adversarial malware generation.	Not mentioned	Devadiga et al. (2023)
	Pre-trained models improved DSLM-GPT2 and GLM-GPT2 detection.	Data set only consisted of a few hundred malware and benign files.	Demirci et al. (2022)
	Demonstrated high ACC and low FPR in the Area Under Curve metric.	The process of extracting sample CFG is very time-consuming.	Gao et al. (2022)
	Archive comparable or even better performance than deep learning baselines by analyzing webpage text content.	ChatGPT sometimes refuses to answer by responding “Unable to determine, need more information”.	Li and Gong (2023)
	High accuracy rates in identifying ransomware.	Employed data set omits older ransomware variants.	Li et al. (2023a)
	Proved the feasibility of pre-trained models for code embedding generation.	Average accuracy score is 7% lower than the best result obtained in the original paper.	García-Soto et al. (2022)

In this part, LLMs can learn existing code fragments to build new security test samples and migrate them to other detection scenarios, optimizing scalability, robustness, and generalization ability of malicious code and malware detection. By the way, most researchers have found that LLMs are worthwhile for detecting and analyzing code and assessing program risks. However, it may be hard and unsuitable for LLM to counterattack threat code or fix vulnerabilities. The relevant work in this section is summarized in [Table 5](#).

3.4. Intrusion and threat discovery in comprehensive security scenarios

Furthermore, an increasing number of researchers are exploring the integration of LLMs into various subtasks of IDS to detect anomalies and events in practical security settings. These subtasks include identifying malicious traffic, analyzing security logs, and detecting potential malicious behaviors. Despite the evolving complexity, stealthiness, and diversity of APTs, LLMs demonstrate superior capabilities in multi-modal information fusion and enhanced detection context. Therefore, our approach focuses on analyzing threat detection holistically rather than solely relying on individual input data objects.

As shown in [Fig. 5](#), considerable effort is dedicated to the development of an intelligent and automated system for detecting anomalies and events. These stages primarily involve the collection and processing of security raw data, extraction of threat features, correlation analysis and matching, and recognition of threats within the target environment. LLMs are evident in nearly every stage of this process.

A.LLMs for Security Data Collection

The scale, type, and complexity of data that needs to be detected vary in different security scenarios. Usually, it is necessary to extract and process a large amount of network traffic data from the detection object before detection, including various protocol packets, log files, and system event information. These data are not only massive in scale but also diverse in type, involving various security threats, ranging from simple malware propagation to APT. LLMs have unique advantages in processing, analyzing, and storing large-scale data, making them widely used in security data collection. For instance, [Setianto et al. \(2021\)](#) developed GPT-2C, specifically for parsing dynamic logs generated by Cowrie SSH honeypots. This model was designed to overcome the limitations of traditional information transformation plugins when dealing with dynamic log topics such as user-generated content. The model was

evaluated using a question-answering task, where the context consisted of a series of utilities and the answers were the utilities predicted by the model. It achieved an inference accuracy of 89% in new domains, with an acceptable execution latency. Data preprocessing and visualization for Cowrie instances, including key metrics like log volume and inference latency, were conducted using Elasticsearch and Kibana dashboards. The simplicity and rapid adaptability of GPT-2C indicate the potential application of LLMs in security raw data processing, Endpoint Detection and Response (EDR), and Security Information and Event Management (SIEM).

[Kholgh and Kostakos \(2023\)](#) introduced an end-to-end PCA-GPT tool, leveraging GPT-3 to generate synthetic network traffic. This synthetic data can substitute real data in tasks like threat modeling, compensating for possible distribution shortcomings in actual datasets and thereby improving the generalization capability of detection systems. The study demonstrated that using GPT-style LLMs to generate network traffic can effectively enhance the robustness and accuracy of threat detection models.

[Deng et al. \(2023\)](#) discovered that LLMs excel in network penetration testing sub-tasks, aiding comprehensive scenario comprehension. They developed PentestGPT, an automatic penetration testing tool that employs sub-modules for different tasks. As the authors note, PentestGPT may become an excellent cybersecurity cognitive engine. PentestGPT's task completion rate surpassed GPT-3.5 by 228.6%, and it gained over 4700 stars on GitHub. By leveraging LLMs, we can achieve more complex and automated cyber threat activities, enabling us to observe and collect more specific security data. Integrated with data collection APIs, PentestGPT has the potential to generate substantial data on attack behaviors through automated testing. This data is valuable for training detection systems in real-world applications, highlighting PentestGPT's potential in security data collection and evolving threat response strategies.

[Ohjuwon et al. \(2023\)](#) invented a patent utilizing LLMs for processing security raw data by inputting prompts into the application programming interface (API) of GPT, facilitating the preprocessing of raw security data. This application enabled the raw data to be transformed into a question-answer format based on the prompts, thus generating original and augmented datasets for fine-tuning models. This method illustrates the key role of LLMs in processing and transforming security raw data, particularly in parsing dynamic and complex logs. The advantage of this approach lies in its ability to quickly

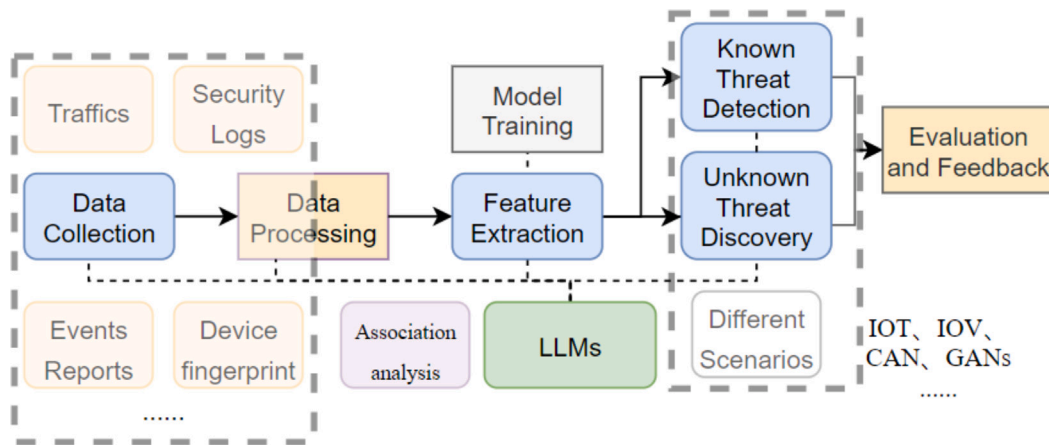


Fig. 5. A diagram about LLMs for Intrusion and Threat Discovery in Comprehensive Security Scenarios.

generate data suitable for fine-tuning, saving time and cost in creating datasets, thereby enhancing the efficiency of processing security data. Furthermore, there exists a plethora of literature on the utilization of LLMs to simulate security data about various security concerns across different scenarios. This body of work serves to enhance the training of detection models by offering comprehensive data supplementation.

In adversarial detection frameworks, there is a correlation between the enhancement of detector performance and generator performance. GPT-style models can learn intricate latent features and produce a substantial volume of synthetic data that may be challenging to replicate in real-world settings. Conversely, BERT-style models excel at capturing contextual information, offering sophisticated structured representation techniques for secure data, thereby enhancing the efficacy of subsequent classifiers. This also indicates that different types of LLMs play different roles in threat detection. Besides, while most LLMs may not be directly applicable to threat detection, their capacity for humanoid reasoning enables them to analyze distinct security concerns, propose solutions for circumventing existing defense mechanisms or protocols, and identify potential attack vectors. Through the validation of diverse attack strategies, the accumulation of extensive data produced by malicious activities can aid in enhancing the detector's capability to identify unfamiliar threats.

B.LLMs for Threat Feature Extraction and Analysis

Using LLMs for threat modeling, threat feature extraction and deep relationship extraction on the obtained security data set can effectively compensate for the shortcomings of traditional threat detection models in this regard. Akbar et al. (2022) utilized the deep semantic learning capacity of the RoBERTa model to infer semantic associations between descriptions of attack techniques and defensive countermeasures. The results indicated that this method could unearth hidden relationships within security data, providing useful inferences and suggestions for security analysis and threat detection with commendable precision. This approach significantly outperformed baseline methods like word2vec and TFIDF, demonstrating the potential of advanced LLMs in enhancing security analysis and threat discovery.

Alam et al. (2022) proposed the LADDER system for extracting attack patterns from CTI to optimize threat detection. LADDER effectively mines complex information within CTI reports using RoBERTa, identifying key features of existing and emerging threats. This assists security analysts and detection systems in preemptively recognizing potential attack methods for proactive defense. Additionally, they contributed a new open-access malware dataset for training future CTI models, further enhancing the capability of threat detection systems.

As for threat modeling, Gadyatskaya and Papuc (2023) utilized GPT-3.5 to automatically create attack trees, a universal method of threat modeling that represents network threats from an attacker's perspective. Their work demonstrated that GPT-3.5 contains sufficient

information about cyber threats and their relationships to generate attack trees, improving the automation level of threat detection and the discovery of unknown threats. However, limitations such as the inability to infer detailed operators were noted.

Boffa et al. (2024) designed the LogPrécis model, which integrates various models including GPT-3 Davinci (175B), BERT, CodeBERT, and CodeBERTa, for automated analysis of text-based Unix shell attack logs. Utilizing the MITRE ATT&CK framework as a guiding structure to capture the causes of attacks, this method automatically learned the attacker's strategy by processing original shell sessions. This approach reduces the number of attack logs and enhances support for attack analysis.

To explore the feasibility of transferring knowledge from large-scale corpora to log-based anomaly detection, Qi et al. (2023) developed LogGPT, leveraging ChatGPT's language interpretation capabilities. This approach integrates log preprocessing, prompt construction, and response parsing. By using the Drain method, log preprocessing extracts structured information from raw logs, grouping them into sequences. Prompt construction involves creating templates with task descriptions, format statements, and human knowledge injections. Response parsing ensures the structured compliance of ChatGPT's responses. The study showed that LogGPT performed well and offered high interpretability on the BGL and Spirit datasets. Compared to the three DL methods, LogGPT provided valuable insights into log-based anomaly detection, despite some performance limitations.

Meanwhile, Lin et al. (2022) proposed a pre-training framework ET-BERT for encrypted traffic classification. This framework utilizes large-scale unlabeled encrypted traffic to learn a universal datagram representation for a series of encrypted traffic classification tasks. ET-BERT was tested in two traffic-specific self-supervised pre-training tasks, such as the masked BURST model and homologous BURST prediction, which capture byte level and BURST level contextual relationships to obtain a universal datagram representation. ET-BERT has strong generalization ability and has achieved new and state-of-the-art performance in five encryption traffic classification tasks, including general encryption application classification, encryption malware classification, VPN encryption traffic classification, Tor encryption application classification, and TLS 1.3 encryption application classification. It is significantly better than existing works by 5.4%, 0.2%, 5.2%, 4.4%, and 10.0%.

Meng et al. (2023) proposed a pre-trained traffic understanding and generation model, NetGPT. By shuffling header fields, segmenting packets in the flow, and introducing various task labels, the training efficiency and effectiveness of downstream tasks such as traffic classification, attack detection, resource scheduling, protocol analysis, and traffic generation have been further optimized. NetGPT also addressed the heterogeneity and diversity of network traffic, as well as

Table 6
Thematic works about LLMs on Intrusion and Threat Discovery in Comprehensive Security Scenarios.

Tasks	Effectiveness	Limitations	literatures
Security Data Collection	Using GPT-2C to parse dynamic logs; Being suitable for EDR and SIEM.	Not Mentioned	Setianto et al. (2021)
	Using GPT-3 to generate composite data of network traffic; compensate for the distribution defects of the data set.	Only two protocols were considered; training data and training duration are limited; CLI.	Kholgh and Kostakos (2023)
	Generate security examples; Assist in automated penetration, adapting to new forms and strategies of threats.	Not Mentioned	Deng et al. (2023)
	Convert original security data into Q&A format; Being suitable for parsing dynamic and complex logs.	Not Mentioned	Ohjuwon et al. 2023 (patent)
Threat Feature Extraction and Analysis	Infer semantic relationship between attack description and defense countermeasures.	Attack and defense information contained in the corpus is limited; Indirectly applied to threat analysis.	Akbar et al. (2022)
	Using LADDER to detect from CTI for attack pattern extraction and trend analysis, threat hunting, and prediction.	Only verified in Android and enterprise networks; Test samples are limited.	Alam et al. (2022)
	Using GPT-3.5 to generate an attack tree automatically, the automation of threat detection is improved.	Cannot reason with refinement operators; Hallucinations of LLMs; lack of checking the integrity of attack tree.	Gadyatskaya and Papuc (2023)
	LogPrécis based on multiple models is used to enhance UNIX shell attack log analysis and feature extraction.	LogPrécis fingerprint may be the same even for different attacks. This design can lead to misclassification.	Boffa et al. (2024)
	LogGPT for log-based anomaly detection with high interpretability.	Model's performance needs further improvement.	Qi et al. (2023)
	ET-BERT for encrypted traffic classification and malicious traffic feature extraction.	Insufficient innovation and modification for BERT.	Lin et al. (2022)
	NetGPT for traffic analysis and generation, optimizing downstream tasks.	no benchmarks and the compared downstream tasks are limited,	Meng et al. (2023)
Real-world threat detection scenarios	Accuracy of intrusion detection using BT-TPF in IOT is over 99%, suitable for resource constrained situations.	Not Mentioned	Wang et al. (2024)
	CAN-SecureBERT and CAN-LLAMA2 for intrusion detection applied in CAN significantly reduce FPR.	LLMs' reasoning speed is limited by computing resources.	Li and Fu (2023)
	HuntGPT anomaly detection system based on ML and XAI. Provide interpretable threat representation.	Achieve lower success in fundamental cyberspace security concepts.	Ali and Kostakos (2023)
	Using SecurityLLM for common threat detection and incident response.	Further fine-tuning is required in specific tasks; continuous updates are required to maintain effectiveness.	Ferrag et al. (2023)

the different dependency relationships of downstream network tasks. Last but not least, through multimodal aggregation and collaborative technology frameworks of large models and small models, LLMs can demonstrate unique advantages in threat detection and comprehensive analysis, which are being introduced in some latest work.

C.LLMs in Real-world Threat Detection Scenarios

In addition to their contributions to security data and threat feature engineering, LLMs can be integrated into actual threat detectors in different network environments to optimize their output.

Wang et al. (2024) proposed a lightweight IoT intrusion detection model named BT-TPF based on BERT-of-Theseus. This model reduces feature dimensions using Siamese networks and employs a large visual transformer as a guide model to teach the smaller Poolformer model. After knowledge distillation, the BT-TPF model's parameter count was reduced by 90%, requiring only 788 parameters. On datasets such as CIC-IDS2017 and TON_IoT, the model achieved an accuracy rate exceeding 99% in traffic-based intrusion detection, significantly outperforming traditional deep learning methods and the latest models, making it particularly suitable for resource-constrained IoT environments.

Li and Fu (2023) developed two new models, CAN-SecureBERT and CAN-LLAMA2, for intrusion detection in Controller Area Networks (CAN). The CAN-LLAMA2 model achieved a balanced accuracy rate

(BA) of 0.999993, with a precision (PREC) rate, a detection rate (DR), an F1 score, and a false alarm rate (FAR) of only $3.1e-6$, which is $1/52$ of the FAR of the most advanced model MTH-IDS. Experiments demonstrated that transformer-based models and LLMs have outstanding performance in CAN intrusion detection, particularly in reducing false positive rates.

Ali and Kostakos (2023) designed HuntGPT, a network anomaly detection system combining machine learning (ML) and explainable AI (XAI), based on GPT-3.5-Turbo and random forest classifiers. On the KDD99 dataset, HuntGPT provided an intuitive user interface and comprehensible threat representation using XAI frameworks like SHAP and Lime. The system demonstrated significant capabilities in cybersecurity knowledge, achieving success rates between 72% to 82.5% in standardized cybersecurity certification exams, including those assessed by Certified Information Security Managers (CISM). This blend of LLM and XAI in a conversational agent effectively enhances the operability and interpretability of the intrusion detection system. The approach offers a program of interpretable, interactive threat detection, reducing manual intervention and improving the efficiency of threat detection and presentation.

As for universal threat detection based on LLMs, Ferrag et al. (2023) successfully fine-tuned SecurityBERT and FalconLLM, leading to the creation of SecurityLLM, a domain-specific LLM for threat detection.

Table 7

Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Literature accessed from superior journals or conferences.	Literature originates from unofficial or unknown sources.
Literature has high correspondence with theme topics.	LLMs were not applied for cyber threat detection.
The experiment is clear and convenient to reproduce.	Experiment relatively unclear or hard to reproduce.
Published in the last three years.	Published too long ago.

In this fixed model, SecurityBERT serves as the threat detection mechanism, while FalconLLM aids in event response and recovery. This approach has shown superior performance in terms of precision and recall on test datasets compared to most previous methodologies. The model can identify 14 different types of threats with an overall accuracy of 98% using various security datasets. However, it requires continuous updates with the latest real datasets to maintain its effectiveness.

As is shown above, LLMs are rarely applied independently for threat detection in most cases, but LLMs can be nested into existing threat detection frameworks for optimization in a certain stage. In this way, not only do LLMs maintain excellent performance on conventional detection indicators, but there is also optimization potential in terms of automation, interpretability, robustness, and generalization ability of threat detection. But not all tasks in different threat detection scenarios, LLMs are better or appropriate. There are still lots of issues being studied. The relevant work in this part is summarized in Table 6.

4. Search methodology

According to the description by Snyder (2019) and other similar literature review methodologies, our reviewing works mainly consist of three phases: (i) review designing, (ii) review conducting, and (iii) result analysis. Each phase was chronologically performed in detail as follows.

4.1. Review designing

To begin with, our reviewing works were guided by customized review rules and research questions.

4.1.1. Review planing

The framework established for investigation delineates a sequence of actions for undertaking the systematic review and meticulously choosing primary references.

I. Scope of study

Since we attempt to explore how to apply LLMs in different scenarios of cyber threat detection, we can not only discuss the direct and independent use of LLMs for threat detection but also research how LLMs are combined with other models. Meanwhile, inclusion and exclusion criteria should be verified according to Table 7.

II. Search Engines and Databases

To systematically collect relevant literature, this study primarily utilized the following five databases: ScienceDirect, Web of Science, SpringerLink, IEEE Xplore, and DBLP. Additionally, search engines such as Google Scholar and Microsoft Academic were employed to supplement the review material. Notably, GPT-4 was utilized for the initial abstract filtering of papers, followed by manual analysis of selected articles to ensure they meet the criteria. Tools like VosViewer were then employed to identify the primary research content within each article.

4.1.2. Research questions

To learn how to apply LLMs in cyber threat detection better, the appropriate and effective selection of research questions is necessary. In this light, four research questions were defined as follows, which were answered in Section 4.3.

- **RQ1.** Compared to traditional models, in which detection tasks do LLMs perform more prominently? The essence of this question is to explore which tasks LLMs are more suitable and focus on which aspect of performance LLMs improved in different tasks.
- **RQ2.** What methodologies were used in each study? We must be concerned about the convenience of relevant works in actual scenarios. Because the effectiveness of LLMs has a relatively harsh demand for data and computing resource environments.
- **RQ3.** What were any limitations that existed in this study? It aims to find if there exists serious prejudice to support for LLMs better or worse.
- **RQ4.** What were the major findings in this study? Apart from the advantages or weaknesses of LLMs in each task, further analysis of the authors' discussion may help us apply LLMs in security products better.

4.2. Review conducting

4.2.1. Search strategy

Usually, the search strategy aims to identify all works associated with primary search terms, incorporating synonyms, abbreviations, and various spelling variations for each term. According to topics, LLMs and cyber threat detection are two keywords. Some keywords associated with LLMs include large language models, GPT, BERT, llama, and transformer. etc. Words associated with cyber threat detection include network traffic, security logs, malware, etc.

Utilizing GPT to identify a set of words strongly linked to the aforementioned topics, we initially formulated a series of retrieval statements. An example of such a statement employed in the Web of Science database is ((Large language model) OR (LLM) OR (GPT) OR (BERT)) AND ((Cyber threat detection) OR (malware detection) OR (networks traffics)), from which additional comparable query sentences were generated.

After collecting sufficient literature, most literature is imported into EndNote, a tool for managing literature. Duplicate literature is subsequently eliminated to compile a selection of literature that is broadly pertinent to thematic works. Subsequently, Python scripts are employed to extract the abstracts of each literature and compile a dictionary collection containing the titles. Through the formulation of precise prompt statements and utilization of the GPT-4 API for automating the content analysis, over 200 research meeting the criteria were initially screened. Then, inclusion within the scope of the review is determined through a manual examination of the abstract and analysis of the article's specifics. Finally, relationship mining tools like Vosviewer can be utilized to uncover the interconnections among various literature topics and pinpoint the security scenarios in which LLMs exhibit exceptional activities.

4.2.2. Selection and clustering

The whole process of our reviewing works has been described in Fig. 6.

As shown above, We employed various search queries across multiple databases and imported the search results into a literature management tool. This process inevitably led to a significant amount of duplicate literature. Following the removal of duplicates, we conducted initial screening on the remaining over 300 research, identifying 215 works on cyber threat detection using LLMs. Subsequently, each article was manually reviewed, with a focus on those meeting specific criteria. Ultimately, 74 research works closely aligned with the topic were

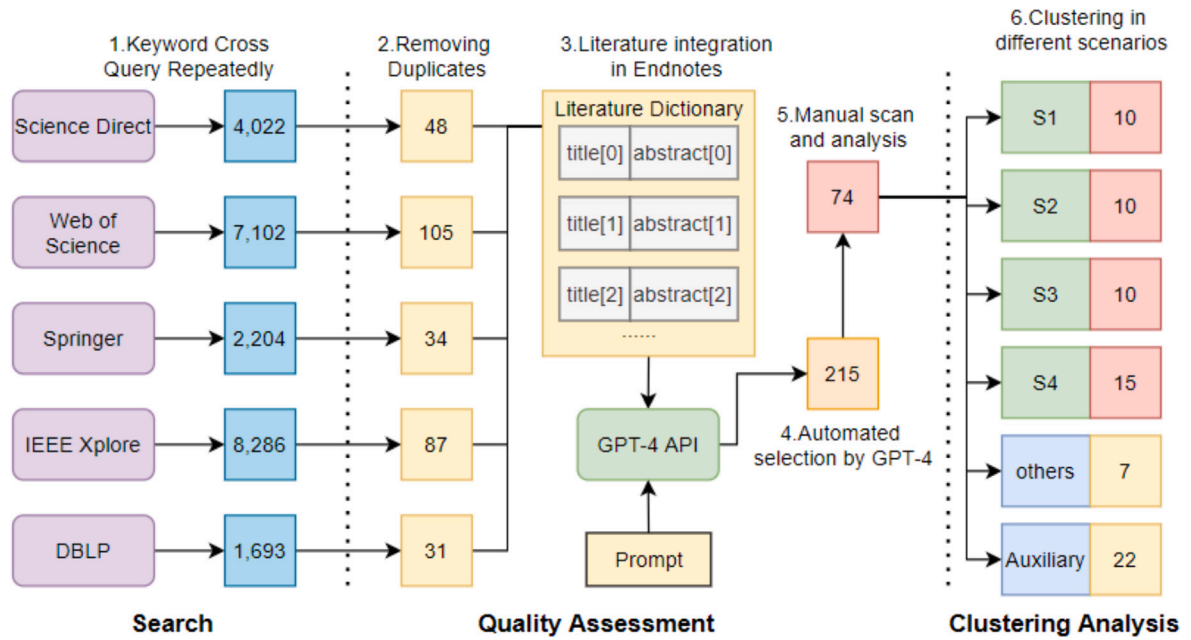


Fig. 6. The process of reviewing works.

4.3. Result analysis

After studying thematic works, we found LLMs are valuable for threat detection in various tasks such as generating heterogeneous raw data, analyzing natural languages or sequences (e.g. phishing threat speech, codes analysis), producing structured reports, etc. This allows us to address the questions in section 4.1.2.

1. LLMs are not always a better choice.

While LLMs may demonstrate superior performance in certain threat detection and discovery tasks, it is important to note that they may not always outperform traditional methods. Some representative assessments of LLMs are summarized in Table 8. We highlighted the varying perspectives on the efficacy of LLMs in different tasks.

From Table 8, we can learn that there exist different voices about the effectiveness of applying LLMs in different tasks. BERT-style models demonstrate exceptional performance in tasks such as NER, RE, and processing structured information or sequences. Their bidirectional design enables them to capture intricate contextual relationships that may be overlooked by conventional models, making them particularly well-suited for in-depth analysis and classification tasks such as phishing email detection, Cyber Threat Intelligence (CTI) analysis, and security log analysis. Conversely, GPT-style models have received more positive assessments for their capacity to generate coherent and contextually relevant text or serialized information. Numerous researchers have opted for GPT-style models to generate adversarial raw data for detection, generate diverse security reports, and simulate attack scenarios for predicting unknown threats, among other applications. Furthermore, GPT-style models possess the capability to incorporate identification, detection, and response functionalities within a system, which is crucial for prompt threat response. GPT-style models are also ideal for enhancing user interfaces and communications in security systems, thus improving the overall efficiency and user experience. Both types of LLMs are capable of effectively analyzing vast amounts of unstructured textual data. By utilizing various types of LLMs in tasks that align with their strengths, the field of cyber threat detection can harness the combined analytical power of these models to effectively identify and mitigate cyber threats.

However, we cannot ignore that the performance of LLMs is unstable due to some factors such as unresolved hallucination issues, the quality of retraining datasets, and the need for domain-specific

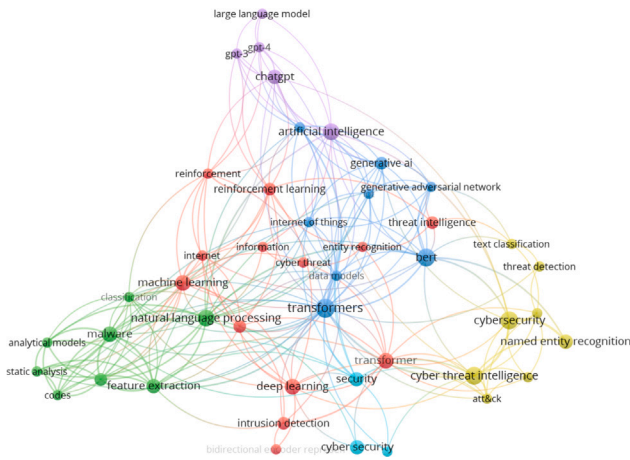


Fig. 7. Keywords clustering in selected thematic works.

identified. Most of these works can be clustered into different threat detection according to manual analysis based on results produced by VOSviewer, as shown in Fig. 7.

In Fig. 7, nodes representing keywords such as intelligence, malware, and natural language processing, particularly those related to threat intrusion, are more prominently displayed, suggesting a greater focus on research in these areas. Consequently, we utilized tools to identify relevant literature and conducted a thorough examination of key content to ascertain the extent to which current research pertains to the utilization of LLMs in this context, as well as the specific applications and roles they play in various tasks. Finally, this review identified 45 articles closely associated with four specific scenarios. Among these, 7 articles were found to be unrelated to the threat detection scenarios discussed, while 22 additional articles served as supplementary material with potential overlap in content, ensuring that redundant information was not included in the main text.

Table 8
Different assessment of LLMs in different tasks.

Tasks	Literature	Authors' Perspectives	Our Assessment
Automatic Detection Reports	Perrina et al. (2023)	High value in RE, CTI, and security reports production.	LLMs are suitable for handling text information from threat detection, generating intelligence or situational reports.
Data Processing for Detection	Kholgh and Kostakos (2023)	Convenient Convenient in generating traffic and other raw data.	LLMs have the potential to become generators for adversarial detection.
SMS Spam Detection	Ghourabi and Alohalay (2023)	High ACC in SMS spam detection, but poor scalability.	LLMs are suitable for directly detecting textual threats but are limited by language and timeliness of training data.
CAN Instruction Detection	Li and Fu (2023)	High ACC and low FPR, but unstable effectiveness.	The stability and AI security issues of LLMs in IDS are worth considering
Threat Modeling and Analysis	Gadyatskaya and Papuc (2023)	Helpful, automation is improved but some problems need solving like hallucinations of LLMs.	LLMs have a certain level of situational understanding ability and potential in threat assessment and prediction.
Malicious Codes Collection	Charan et al. (2023)	Helpful, accelerate generating malicious codes, but sometimes reluctant in corresponding.	LLMs can help simulate complex attacks and defenses, during which more security data can be generated.
Vulnerability analysis in CTI	Liu et al. (2023)	Limited, ChatGPT still cannot be used directly for VDM tasks.	When dealing with highly specialized intelligence, we should construct domain-specific LLMs rather than use generic LLMs directly.

knowledge. On the contrary, traditional models may be better suited for real-time intrusion detection, threat monitoring, and low-level threat analysis due to their lower latency and more targeted approach. These models are typically designed to handle specific types of data and threats, making them more effective for real-time or single-task detection and response. Furthermore, traditional models especially ML models, offer greater transparency and ease of interpretation in their decision-making processes, a critical aspect in security contexts. However, the intricate landscape of cyberspace, including the most advanced deep learning methods, may not effectively detect threats. In cases where valuable assets are at stake, Local Linear Models (LLMs) may prove to be indispensable.

While these findings may seem to contradict those of other researchers, they are not necessarily incongruent. It is generally deemed inappropriate to solely rely on LLMs for cyber threat detection, as indicated by numerous studies. Instead, integrating various methods, such as incorporating LLMs at specific stages, is considered a more effective approach to optimizing the detection scheme.

II. LLMs are mostly applied indirectly in research.

According to Section 3, it is an indispensable and more popular option of integrative methods, combining the strengths of both LLMs and traditional models. Most of the existing research was conducted jointly with GPT, BERT, and ML, DL methods. However, a segment of researchers have still innovated by developing their domain-specific LLMs for threat detection or other tasks. While GPT series LLMs were applied more in generating various data of traffic, codes, test samples, or other data, BERT series LLMs were more adapted in analysis works like RE and NER. Simultaneously, we found the domain-specific LLMs fine-tuned on GPT-2 or other previous open-source LLMs, are more favored. The reason could be there exist restrictions to access or retrain GPT-3 or later closed-source LLMs. Besides, some researchers have critically assessed the advantages and disadvantages of LLMs by comparative experiments. Utilizing LLMs directly for threat detection may be suitable only on a few occasions like phishing E-mail detection and codes static analysis etc. This is mainly because LLMs were originally designed for natural language processing, excelling in text and sequence-related tasks. However, for advanced threat detection tasks such as steganographic component detection in multimodal information and complex security log analysis, LLMs often require collaboration with other technologies. We believe that in the future, LLMs

will develop enhanced emergent capabilities, providing opportunities to overcome current limitations.

III. There are still some limitations in this research.

Unfortunately, not all studies articulated fully detailed methods of how they develop models or shared datasets for readers to retrain local LLMs. We know the actual effectiveness of LLMs can be influenced by datasets and knowledge dramatically. Next, there are too few datasets involved in most comparative experiments, so we cannot confirm whether LLMs can truly maintain the same performance on the vast majority of datasets. To verify the effectiveness of LLMs, more research needs to be conducted. Last, nearly all researchers have attempted to demonstrate the feasibility of LLM in specific tasks rather than multi-task scenarios. Scenarios in which LLMs and small models work together for threat detection have not been discussed in more detail. Consequently, there is a lack of unified standards or datasets to guide the implementation of LLMs in actual security systems. New methods and datasets for evaluating the effectiveness of detection models should be designed, including consideration of detection costs.

IV. Further research is warranted on LLM-driven methods.

In summary, our analysis of existing research indicates that LLMs demonstrate significant application in the realm of cyber threat detection, particularly in the following areas: Firstly, LLMs serve as a pertinent tool for text processing and analysis in the realm of cyber threat detection. Specifically, LLMs modeled after BERT exhibit efficacy in automating precise threat text analysis to discern the potentially harmful nature of network activities. Additionally, LLMs can produce diverse structured intelligence and situational analysis reports, thereby enhancing the interactive nature of threat detection and the comprehensibility of outcomes. Secondly, LLMs demonstrate proficiency in recognizing named entities within unstructured data, uncovering latent relationships among disparate entities, and facilitating the generation of structured content. Consequently, LLMs play a beneficial role in facilitating information sharing among security devices manufactured by various companies. LLMs have the potential to serve as an effective tool for enhancing cyber security situational awareness and comprehension, thereby facilitating the amalgamation of identification, detection, and response mechanisms, ultimately reducing threat detection time and enhancing detection scalability. In contrast to traditional models, LLMs exhibit an enhanced capacity for processing extensive amounts of information simultaneously and possess a broader context window. Consequently, the integration of LLMs with technologies such as big

data security and intelligence warning systems presents an appealing prospect. Thirdly, integrating LLMs with conventional methods has the potential to mitigate the occurrence of FPR and FNR in cyber threat detection. On one hand, LLMs can effectively conduct log correlation analysis to eliminate extraneous information within the given context. Furthermore, they can aid in noise reduction by sifting through numerous alarm events and identifying those necessitating manual scrutiny. This approach not only diminishes the false alarm rate but also minimizes the expenses associated with manual analysis. On the other hand, LLMs can analyze threat data and log information gathered from various sources, enabling the identification of previously undetected threat activities spanning multiple dimensions and aiding in the detection of some unknown cyber threats. Lastly, LLMs have the potential to serve as effective generators for detecting adversarial threats by generating diverse security samples and attack vectors, thereby enhancing data quality and reducing acquisition costs. Specifically, GPT-style LLMs possess superior content generation capabilities that are lacking in conventional models. Leveraging these LLMs can augment the variety of training data and produce adversarial samples based on specified prompts. For instance, employing GPT to devise novel attack strategies against established defense mechanisms and conduct verification processes can aid in uncovering additional vulnerabilities and new threats.

Simultaneously, scholarly research has identified constraints in the practical implementation of LLMs, including the impact of factors such as data size, quality, structure, and timeliness on the stability of detection outcomes. The inherent security vulnerabilities in AI technology may result in negative outcomes such as data breaches and diminished detection efficacy. Moreover, detection techniques reliant on a limited number of LLMs may prove ineffective against injection attacks. The primary utility of LLMs lies in their intellectual emergence, with the efficacy of LLMs typically contingent upon the scale of the model and the volume of data utilized. It is important to acknowledge that LLMs enhance information processing to a certain degree and serve as a novel tool for data aggregation, yet their ability to acquire new information beyond the confines of the training dataset remains limited.

Despite some deficiencies in existing works, the vast majority of research indicates that LLMs have enormous potential in cyber threat detection. There are lots of optimizations needed, especially how to provide high-quality retraining data for LLMs, how to make LLMs' performance more stable, and how to decrease the cost of LLMs in real-world systems. Thus, more and more researchers are joining this field to explore a better solution or technical framework with LLMs for threat detection and discovery, especially cooperation between LLMs and traditional models.

5. Discussion

A. Summary

In this study, we found LLMs were applied in quite a few tasks of cyber threat detection, summarizing four security scenarios highly related to that. Thematic research has shown that LLMs have high potential in tasks such as threat intelligence processing, textual threat detection, and static code detection, which is consistent with the conclusions of some mainstream technology reports. Some highly relevant thematic work with comprehensive analysis may help security workers build domain-specific LLMs for cyber threat detection in specific scenarios.

The innovation of this review lies in its approach to addressing current problems in threat detection, retrospectively tracing the improvements made by LLMs in different specific security scenarios, rather than sequentially listing these works categorized by types of input data. The main reason is that we found not all aspects of threat detection benefit from LLMs. Meanwhile, it does not effectively highlight where LLMs are most valuable to discuss the combination of all threat detection tasks with LLMs.

However, there still exist some limitations in the review. For example, all thematic works presented are mainly sourced from literature databases within three years, without further expanding the scope to include more patents, products, books, white papers, and organizational reports. Additionally, the organization of the review is more focused on the benefits of the logic perspective, lacking more detailed analysis and related works on specific threat detection tasks, such as traffic detection and log analysis.

B. More Topics

Besides, it is worth talking more issues related to topics, most of which are also discussed by researchers these days.

I. The contradiction between LLMs and threats.

LLM, one of the most advanced AI technology tools, is a double-edged sword for cyberspace security. It is argued that more threats can be disposed of by LLM while more threats are generated by LLM. Specifically, LLM reduces the threshold for hackers and the process of attack-chain faster without restrictions. In contrast, LLM can provide a series of supports to address these new problems and shorten the response chain of threats.

II. The relationship between LLM and cyber security.

There exists a complementary relationship between the LLM and cyberspace security. On one hand, security for LLM, which is dedicated to securing LLMs, is an important prerequisite for achieving secure applications of LLM. On the other hand, LLM for security, which is dedicated to using LLM to solve security problems, can promote the resolution of LLM's security.

III. Which one is better, efficiency or performance?

In the practical optimization of LLM, it is a pivotal dilemma whether a larger LLM is inherently better or efficiency holds greater value. Generally, larger LLMs imply advanced capabilities and improved effectiveness, yet concurrently incur higher temporal expenditure as well as costs for training and operation. Conversely, heightened efficiency allows LLMs to respond more swiftly to tasks, striving to approach the upper limit of performance in their current size as closely as possible. Optimizing for both size and efficiency simultaneously is often not feasible, necessitating a tailored approach that aligns with the specific requirements of the detection tasks at hand.

IV. Is LLM for security trustworthy?

It was reported that users and vendors may encounter issues such as distrust and conflicts of interest (Gupta et al., 2023). The "Q&A alignment problem" refers to the situation where users cannot always get what they want due to the restriction of output. And some countries even think LLM from other regions may threaten their security or help the locals break laws, and have strictly prohibited products based on LLM. So, is it trustworthy to use LLM to detect various threats without any side effects?

All of these similar problems are being discussed these days in different articles, reports, and even newspapers. It is not only a technical issue of how to integrate LLMs, threat detection, and cyberspace security more efficiently.

C. Future Works

Based on existing findings, it is advisable for future studies within this field to investigate the following areas:

I. Improvements in existing directions

It is a considerable option to study the potential applications of LLMs in security data processing, enhancements to existing Traffic-GPTs or Log-GPTs, strategies for mitigating noise in extensive log alerts, identification of linguistic threat vectors and serialized data, development of threat pattern recognition systems, methods for detecting adversarial activities, and the generation of structured reports.

Meanwhile, it makes sense to explore the integration of LLMs with advanced DL frameworks.

II. Exploring Emerging Capabilities of LLMs

Further research should be conducted on the utilization of LLMs for internal threat and espionage detection, as well as the development of acceptable security detection reasoning frameworks, threat modeling and prediction, and comprehensive analysis of various threat carriers concurrently, including traffic, logs, and software. Besides, it is important to acknowledge that LLMs have the potential to not only detect threats but also aid in their mitigation. This capability could advance cybersecurity posture awareness to a higher level of understanding, potentially introducing new definitions and entirely novel technological frameworks for cyber threat detection.

III. More Interdisciplinary Crossover

Given LLMs' strengths in deep language understanding and human-like expression, there is potential for their integration with cyber cognitive threat detection, hackers' psychology, and sophisticated game theory to bolster the scope and efficacy of threat detection efforts. For instance, it is evident in numerous APT reports, such as the 2021 Microsoft Exchange Server Zero Day Vulnerability Attack, that perpetrators have employed persistent low-level attack vectors to obfuscate their true attack vectors. Consequently, employing LLMs to analyze the deceptive tactics utilized by threat actors in APT reports can facilitate enhanced identification of APT groups and forecasting of forthcoming attack patterns. These tasks heavily depend on cognitive and emotional capabilities akin to those of humans, which are challenging to replicate in conventional models.

IV. Challenges of Domain-Specific LLMs

It is imperative to address the security concerns, ethical considerations, and privacy protection challenges associated with LLMs. Furthermore, enhancing the interpretability of LLMs in threat detection is a shared focus for both security professionals and artificial intelligence researchers.

We plan to delve deeper into some areas above, conducting research works to find more applications of LLMs on cyber threat detection. For example, our current research project incorporates multiple essential elements. Initially, the implementation of a model resembling ET-BERT for refining network traffic format is intended to improve the efficacy of downstream classifiers in traffic categorization. Subsequently, a model akin to Log-GPT will be developed for the thorough examination of network behavior. Finally, the incorporation of a domain-specific LLM as a security framework allows for the aggregation, integration, and analysis of security data from various origins, facilitating the establishment of a unified mechanism for comprehensive multi-channel detection.

6. Conclusion

In this review, we have systematically explored a brief history of LLMs and cyberspace threat detection, analyzing the potential of LLMs in addressing existing challenges in threat detection. By investigating relevant literature over the past three years, we discussed applications of LLMs in different tasks of typical security scenarios, enhancing the adaptability and automation level of threat detection. In the context of this, we summarized the current development status in this interdisciplinary field and explored the potential works and related topics of further LLM applications. This review focuses on providing a comprehensive perspective on the use of LLMs in threat detection, offering valuable insights for researchers interested in designing and optimizing threat detection using LLMs.

CRedit authorship contribution statement

Yiren Chen: Writing – original draft, Visualization, Methodology, Investigation. **Mengjiao Cui:** Investigation, Data curation. **Ding Wang:** Investigation. **Yiyang Cao:** Investigation. **Peian Yang:** Writing – review & editing, Validation, Supervision. **Bo Jiang:** Writing – review & editing, Resources, Funding acquisition, Formal analysis. **Zhigang Lu:** Formal analysis, Conceptualization. **Baoxu Liu:** Supervision, Formal analysis.

Declaration of competing interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

Data availability

No data was used for the research described in the article.

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